SOWUNDARYA R

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Summary

A self-driven and passionate Computer Science and Engineering student with a strong foundation in Data Structures, Algorithms, and web technologies. Seeking an entry-level opportunity to apply problem-solving skills and build scalable, user-friendly interfaces while continuously learning and growing in a collaborative environment.

Education

2023–2027	**BANNARI AMMAN INSTITUTE OF TECHNOLOGY** B.E - Computer Science and Engineering CGPA: 8.24 (upto 5th semester)
2022–2023	**S.V.N Matric Higher Secondary School** 12th - 96.16%
2020–2021	**S.V.N Matric Higher Secondary School** 10th - Pass

Skills

Languages	: Java, SQL
Web Technologies	: React, Node.js, Express.js, HTML5, CSS3, Bootstrap, Javascript
Databases	: MongoDB, MySQL
Concepts	: Data Structures and Algorithms (DSA), OOPS, DBMS
Tools	: Git, Postman, VS Code

Projects

MAP-BASED NEWS RETRIEVAL SYSTEM

Technologies: React.js, Node.js, Express.js, PostgreSQL, RESTful APIs, Leaflet.js, External News APIs

- Developed a full-stack web application that displays location-based news on an interactive map interface.
- Designed a responsive frontend using React.js and integrated it with a secure Node.js and Express.js backend connected to a PostgreSQL database.
- Implemented RESTful APIs for user authentication, data retrieval, and CRUD operations, enabling real-time news updates based on user location.
- Ensured smooth communication between the frontend, backend, and external news APIs for an efficient and user-friendly experience.

WATER QUALITY PREDICTION

Technologies: Python, Scikit-learn, HTML , CSS, Tailwind CSS, Folium, GeoPandas, Flask, RESTful APIs

- Developed a machine learning-based web application to predict the potability of water and visualize results through interactive geospatial maps.
- Implemented Random Forest and Gradient Boosting models for accurate classification, evaluated using ROC-AUC and accuracy metrics.
- Designed a responsive frontend using React and Tailwind CSS for real-time result visualization and user interaction.
- Integrated APIs for smooth data exchange between frontend and backend, and used Folium and GeoPandas for regional potability visualization.

Certificates & Achievements

CONFERENCE	Presented a research paper titled "Music Recommendation Based On Facial Expression" at the 4th National Conference on Recent Advancements in Science, Engineering and Technologies (RASET-2024), Bannari Amman Institute of Technology. Received the Best Paper Award for outstanding innovation.
CERTIFICATION	Completed the NPTEL course **"Programming in Java"** with an outstanding score of **94% (Elite)**.

Competitions

Imaginative 2.0	The Ultimate Prodman Challenge, Empyrean Annual Fest, Indian Institute of Management (IIM), Jammu (2024).
Hackathon	Participated in **TANCAM Women Hackathon 2024** under the theme "Water Quality Prediction using Machine Learning". Developed an ML-based model to predict water potability and visualized results using geospatial data.